

64th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION

Pooled Fecal Allogenic Microbiotherapy for Refractory Gastrointestinal Acute Graft-Versus-Host Disease: Results from the Early Access Program in France

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Abstract Introduction Acute graft-versus-host disease (aGVHD) is a major source of mortality following allogeneic hematopoietic cell transplantation (allo-HCT). Standard 1st line therapy for the treatment of aGVHD involves corticosteroids (CS). However, more than 50% of patients are refractory to CS (SR-aGVHD) and the associated mortality rate is of up to 80%. Recently, Ruxolitinib has been approved as 2nd line treatment for SR-aGVHD. However, patients with severe gastrointestinal (GI) aGVHD seem less likely to respond to Ruxolitinib and have poor outcomes with limited therapeutic options.

In this context, fecal microbiotherapy has shown promising results in several pilot studies in patients with refractory GI-aGvHD. Here we report clinical outcomes from 81 patients diagnosed with SR or dependent (SD) GI-aGvHD treated with the pooled allogeneic microbiotherapy MaaT013 as part of the Early Access Program (EAP) in France.

Methods 81 patients with SR/SD GI-aGvHD (Classical n=62, late onset n=12, overlap syndrome n=7) were treated with MaaT013 as part of an EAP in France. These patients had failed 1 to 6 prior systemic GvHD treatment lines (median 2; 66/81 received Ruxolitinib). Most patients had grade III-IV aGvHD (11% grade II, 51% grade III, 38% grade IV).

For each patient, a total of 3 MaaT013 administrations were planned every 7 +/- 2 days. Each dose comprised 30g of feces in 150 mL of diluent, from 4 to 8 healthy donors, administered by enema (except for 1 patient by nasogastric tube).

Treatment response was calculated among all treated patients based on aGvHD staging and grading at day 28 (D28) at the time of the EAP request.

Results At D28, the GI-overall response rate (ORR) was 56%: 30 complete response (CR, 37%), 11 very good response rate (VGPR, 14%), 4 partial response (PR, 5%). The GI-ORR was higher in patients with lower grade aGvHD (89% in grade II, 66% in grade III, 32% in grade IV) and higher in SD versus SR patients (92% versus 49%). Including skin and liver symptoms (n=78), response rate was 49%, including 24 CR, 11 VGPR and 3 PR and inversely correlated with aGvHD initial grade (88% in grade II, 55% in grade III, 30% in grade IV).

Overall survival (OS) was 51% at 6 months (M6) and 39% at M12. The median follow-up among surviving patients was 355 days (range, 53-731). OS was significantly higher in patients achieving at least GI-PR at D28 (Responder, R; n=45) compared to patients in treatment failure (Non-responder, NR; n=35): 69% versus 28% at M6, 59% versus 14% at M12, respectively. Median survival duration in R was 451 days versus 32 days in NR.

Interestingly, aGvHD response was improved in the subgroup of 31 patients previously treated with Ruxolitinib as 2nd line and MaaT013 as 3rd line (65% D28 GI-ORR, OS M6 55% and 49% M12 and 74% versus 15% at M6 and 74% versus 0% at M12 for R and NR respectively).

MaaT013 displayed a good overall safety profile in the EAP population: 20 pharmacovigilance cases were reported in 18 patients: sepsis in 11 patients, *C. difficile* colitis in 2, *E. coli* osteoarthritis in 1, *G. silvicola* in stools from 1, *P. aeruginosa* sinusitis in 1, appearance of air bubbles in the mesorectum in 1, respiratory distress in 1. No pathogen transmission was reported. In 2 patients, non-pathogenic commensal bacteria isolated following infectious events were detected in the administered MaaT013 batch. Causality could not be formally excluded in these cases.

The overall incidence of bacteremia (14%) remains low, compared to an incidence of 31% to 74% in bloodstream infections reported in patients with GI-aGvHD. This suggests that fecal microbiotherapy may have a protective effect on bacterial translocation, but this needs to be confirmed in further clinical trials.

47 deaths have been reported; the cause of which was GvHD in 21 patients, severe infection in 13, relapse of underlying malignancy in 6, COVID-19 in 3, hemorrhage during surgery in 1, neurological complications post allo-HCT in 1, and cardiac arrest in 2 patients. No causality link with MaaT013 administration has been identified.

Conclusion Overall, EAP clinical data showed that MaaT013 was safe and effective for the treatment of SR/SD-GI-aGvHD especially in patients having previously received ruxolitinib. Interestingly, GI-response to aGvHD correlates with increased OS, suggesting a strong favorable benefit-risk profile for MaaT013. A Phase 3 trial is currently ongoing to confirm these results in ruxolitinib-refractory patients (NCT04769895).

Disclosures **Malard:** Therakos/Mallinckrodt: Honoraria; Sanofi: Honoraria; Astellas: Honoraria; JAZZ pharmaceuticals: Honoraria; Biocodex: Honoraria; Janssen: Honoraria; Takeda: Honoraria; Novartis: Honoraria; Gilead: Honoraria; Celgene-BMS: Honoraria. **Chevallier:** Pfizer: Research Funding; Incyte: Research Funding; Abbvie: Honoraria; Takeda: Honoraria; Jazz Pharmaceuticals: Honoraria. **Beauvais:** Gilead: Honoraria. **Bruelle:** MaaT Pharma: Current Employment. **Plantamura:** MaaT Pharma: Current Employment. **Mohty:** Jazz Pharmaceuticals: Honoraria, Research Funding; Celgene: Honoraria; Bristol Myers Squibb: Honoraria; Takeda: Honoraria; Novartis: Honoraria; Amgen: Honoraria; Astellas: Honoraria; Adaptive Biotechnologies: Honoraria; Oncopeptides: Honoraria; Pfizer: Honoraria; GSK: Honoraria; Sanofi: Honoraria, Research Funding; Janssen: Honoraria, Research Funding; Gilead: Honoraria.

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Baseline characteristics and efficacy results of treated EAP patients

Baseline characteristics and clinical results of treated 2nd patients		Treated patients (N=81)	
Patient characteristics			
Gender, n (%)	Male	40 (49%)	
	Female	41 (51%)	
Age at first MasT013 administration (years)	Median [range]	58 [18;73]	
Time between aGVHD diagnosis and first MasT013 dose, days	Median [range]	33 [8;440]	
Number of previous lines of treatment, n	Median [range]	2 [1;6]	
Steroid status	Steroid refractory-aGVHD	68 (84%)	
	Steroid dependent-aGVHD	13 (16%)	
Type of aGVHD	Classical	62 (77%)	
	Late onset	12 (15%)	
	Overlap syndrome	7 (9%)	
aGVHD grading (MAGIC), n (%)	I	0	
	II	9 (11%)	
	III	41 (51%)	
	IV	31 (38%)	
GVHD organ involvement at EAP inclusion	GI only	48 (59%)	
	GI + skin	19 (23%)	
	GI + liver	3 (4%)	
	GI + skin + liver	3 (4%)	
	Missing data for skin and liver	8 (10%)	
Stage skin aGVHD	Stage 0	53 (65%)	
	Stage 1	13 (16%)	
	Stage 2	6 (7%)	
	Stage 3	3 (4%)	
	Stage 4	0	
	Missing data	6 (7%)	
Stage liver aGVHD	Stage 0	67 (83%)	
	Stage 1	4 (5%)	
	Stage 2	2 (2%)	
	Stage 3	0	
	Stage 4	0	
	Missing data	8 (10%)	
Stage gut aGVHD	Stage 0	0	
	Stage 1	9 (11%)	
	Stage 2	18 (22%)	
	Stage 3	23 (28%)	
	Stage 4	31 (38%)	
Results			
GI response at D28		All patients (n=81)	Ruxolitinib refractory patients who received MasT013 as 3 rd line (N=31)
	ORR, n (%)	45 (56%)	20 (65%)
	CR, n (%)	30 (37%)	19 (61%)
	VGPR, n (%)	11 (14%)	1 (3%)
	PR, n (%)	4 (5%)	0
Overall survival		All patients (n=80)	Ruxolitinib refractory patients who received MasT013 as 3 rd line (N=30)
	M6		
	All patients	51%	55%
	Responders	69%	74%
	Non-responders	28%	13%
	M12		
	All patients	39%	49%
	Responders	59%	74%
	Non-responders	14%	0%

Abbreviations: GI: gastrointestinal; aGVHD: acute graft-versus-host-disease; M: month ; D: day; ORR: overall response rate; CR: complete response; VGPR: very good partial response; PR: partial response

Figure 1.